

Memory Interfacing: RAM & ROM Decoding -- MPCA Module 4

Definition: Memory Interfacing is the electronic hardware arrangement that connects semiconductor memory chips (RAM and EPROM/ROM) to the microprocessor's Address, Data, and Control buses using Address Decoding Logic to assign unique physical address ranges to each memory chip.

1. Detailed Technical Explanation

1. 8086 Memory Organization Review

- * The 8086 microprocessor has a 20-bit Address Bus (A₀ - A₁₉), enabling it to address up to 1 MB (2²⁰ bytes) of physical memory.

Microprocessor 8086		Memory Banks (1 MB Total)	
+-----+		+-----+	
	A0 = 0		EVEN BANK (512 KB RAM/ROM)
Data Lines D0-D7	=====>		Data Lines D0-D7
	/BHE = 0		ODD BANK (512 KB RAM/ROM)
Data Lines D8-D15	=====>		Data Lines D8-D15
+-----+		+-----+	

2. Address Decoding Logic (74LS138 3-to-8 Decoder)

The 74LS138 decoder is the industry standard IC used to generate chip select (CS) signals for multiple memory chips.

74LS138 3-to-8 DECODER			
+-----+			
A19 ----->	G1 (Active High Enable)		
A18 ----->	/G2A (Active Low)		
M//IO ----->	/G2B (Active Low)		
A17 ----->	C (Select Input 2)	----->	/Y0 (CS for Chip 0)
A16 ----->	B (Select Input 1)	----->	/Y1 (CS for Chip 1)
A15 ----->	A (Select Input 0)	----->	/Y2 (CS for Chip 2)
		----->	...
+-----+			

Memory Map Design Example:

To interface a 64 KB EPROM chip (Address lines A₀ - A₁₅ used internally inside chip):

3. Core Concepts & Memory Keywords

- * Address Decoding: Translating upper address bus bits into unique CS chip select signals.

4. Must-Write Points for Exams

- * ROM/EPROM is always placed at the top of memory (FFFF0H) because 8086 starts execution at FFFF0H upon reset.

5. Quick Recall Flow

20-bit Address Bus -> Lower bits (A0-Ak) to Memory Chip -> Upper bits to 74LS138 Decoder -> Chip Select /CS Generated